In [ ]:

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'''>>>>> start CodeP2.3F25
    V.P. Carey ME249, Fall 2025

Intro to Neural Network Modeling
Data arrays for hybrid solar/fossil-fuel gas turbine power system'''

'''
#create input data array, normalizing input temp
#T1(K), gamma, , qsol(kW):
xdata = []
xdata =  [[ 318.0 , 0.0 , 500.0 ], [ 318.0 , 0.0 , 1000.0 ]]
xdata.append([ 318.0 , 0.0 , 1500.0 ])
xdata.append([ 318.0 , 0.0 , 2000.0 ])
xdata.append([ 318.0 , 0.0 , 2500.0 ])
xdata.append([ 318.0 , 0.25 , 500.0 ])
xdata.append([ 318.0 , 0.25 , 1000.0 ])
xdata.append([ 318.0 , 0.25 , 1500.0 ])
xdata.append([ 318.0 , 0.25 , 2000.0 ])
xdata.append([ 318.0 , 0.25 , 2500.0 ])
xdata.append([ 318.0 , 0.5 , 500.0 ])
xdata.append([ 318.0 , 0.5 , 1000.0 ])
xdata.append([ 318.0 , 0.5 , 1500.0 ])
xdata.append([ 318.0 , 0.5 , 2000.0 ])
xdata.append([ 318.0 , 0.5 , 2501.0 ])

xdata.append([ 303.0 , 0.0 , 500.0 ])
xdata.append([ 303.0 , 0.0 , 1000.0 ])
xdata.append([ 303.0 , 0.0 , 1500.0 ])
xdata.append([ 303.0 , 0.0 , 2000.0 ])
xdata.append([ 303.0 , 0.0 , 2500.0 ])
xdata.append([ 303.0 , 0.25 , 500.0 ])
xdata.append([ 303.0 , 0.25 , 1000.0 ])
xdata.append([ 303.0 , 0.25 , 1500.0 ])
xdata.append([ 303.0 , 0.25 , 2000.0 ])
xdata.append([ 303.0 , 0.25 , 2500.0 ])
xdata.append([ 303.0 , 0.5 , 500.0 ])
xdata.append([ 303.0 , 0.5 , 1000.0 ])
xdata.append([ 303.0 , 0.5 , 1500.0 ])
xdata.append([ 303.0 , 0.5 , 2000.0 ])
xdata.append([ 303.0 , 0.5 , 2500.0 ])

xdata.append([ 288.0 , 0.0 , 500.0 ])
xdata.append([ 288.0 , 0.0 , 1000.0 ])
xdata.append([ 288.0 , 0.0 , 1500.0 ])
xdata.append([ 288.0 , 0.0 , 2000.0 ])
xdata.append([ 288.0 , 0.0 , 2500.0 ])
xdata.append([ 288.0 , 0.25 , 500.0 ])
xdata.append([ 288.0 , 0.25 , 1000.0 ])
xdata.append([ 288.0 , 0.25 , 1500.0 ])
xdata.append([ 288.0 , 0.25 , 2000.0 ])
xdata.append([ 288.0 , 0.25 , 2500.0 ])
xdata.append([ 288.0 , 0.51 , 500.0 ])
xdata.append([ 288.0 , 0.5 , 1000.0 ])
xdata.append([ 288.0 , 0.5 , 1500.0 ])
xdata.append([ 288.0 , 0.5 , 2000.0 ])
xdata.append([ 288.0 , 0.5 , 2500.0 ])

xdata.append([ 268.0 , 0.0 , 500.0 ])
xdata.append([ 268.0 , 0.0 , 1000.0 ])
xdata.append([ 268.0 , 0.0 , 1500.0 ])
xdata.append([ 268.0 , 0.0 , 2000.0 ])
xdata.append([ 268.0 , 0.0 , 2500.0 ])
xdata.append([ 268.0 , 0.25 , 500.0 ])
xdata.append([ 268.0 , 0.25 , 1000.0 ])
xdata.append([ 268.0 , 0.25 , 1500.0 ])
xdata.append([ 268.0 , 0.25 , 2000.0 ])
xdata.append([ 268.0 , 0.25 , 2500.0 ])
xdata.append([ 268.0 , 0.5 , 500.0 ])
xdata.append([ 268.0 , 0.5 , 1000.0 ])
xdata.append([ 268.0 , 0.5 , 1500.0 ])
xdata.append([ 268.0 , 0.5 , 2000.0 ])
xdata.append([ 268.0 , 0.5 , 2500.0 ])
  '''

'''
#outputs
#alpha = air/fuel ratio , cycle efficiency:
ydata =  [[ 35.1316 , 0.3808 ],[ 40.3764 , 0.38686 ]]
ydata.append([ 47.4620 , 0.3930 ])
ydata.append([ 57.5639 , 0.39949 ])
ydata.append([ 73.1286 , 0.40612 ])
ydata.append([ 49.1110 , 0.4023 ])
ydata.append([ 56.4428 , 0.40605 ])
ydata.append([ 66.3479 , 0.4098 ])
ydata.append([ 80.4695 , 0.4133])
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ydata.append([ 102.2276 , 0.4175 ])
ydata.append([ 63.0904 , 0.41540 ])
ydata.append([ 72.5092 , 0.4175 ])
ydata.append([ 85.2338, 0.4197 ])
ydata.append([ 103.3750 , 0.42192 ])
ydata.append([ 131.3266 , 0.4242 ])

ydata.append([ 34.273 , 0.3952 ])
ydata.append([ 38.99026 , 0.4012 ])
ydata.append([ 45.2133, 0.4073 ])
ydata.append([ 53.8000 , 0.4136 ])
ydata.append([ 66.4130 , 0.4201 ])
ydata.append([ 47.922 , 0.4178 ])
ydata.append([ 54.518 , 0.4215 ])
ydata.append([ 63.220 , 0.4252 ])
ydata.append([ 75.226 , 0.4290 ])
ydata.append([ 92.862 , 0.4329 ])
ydata.append([ 61.572 , 0.4315 ])
ydata.append([ 70.0468 , 0.43373 ])
ydata.append([ 81.226 , 0.43597 ])
ydata.append([ 96.653 , 0.4382 ])
ydata.append([ 119.3124 , 0.44045 ])

ydata.append([ 33.4521 , 0.40913 ])
ydata.append([ 37.6911, 0.4150 ])
ydata.append([ 43.1602 , 0.4209 ])
ydata.append([ 50.4858 , 0.4271 ])
ydata.append([ 60.8067 , 0.4334 ])
ydata.append([ 46.7865 , 0.4328 ])
ydata.append([ 52.7151 , 0.43646 ])
ydata.append([ 60.36425 , 0.44016 ])
ydata.append([ 70.6099 , 0.443926 ])
ydata.append([ 85.0447 , 0.4477 ])
ydata.append([ 60.1208 , 0.44721 ])
ydata.append([ 67.7391 , 0.44940 ])
ydata.append([ 77.56830 , 0.4516 ])
ydata.append([ 90.73410 , 0.4538 ])
ydata.append([ 109.2828 , 0.4560 ])

ydata.append([ 32.4123 , 0.42694 ])
ydata.append([ 36.0807 , 0.4325 ])
ydata.append([ 40.6854 , 0.4383 ])
ydata.append([ 46.6374 , 0.4442 ])
ydata.append([ 54.6293 , 0.4503 ])
ydata.append([ 45.3472 , 0.4519 ])
ydata.append([ 50.4796 , 0.4555 ])
ydata.append([ 56.9219 , 0.4591 ])
ydata.append([ 65.2492 , 0.4628 ])
ydata.append([ 76.4304 , 0.4665 ])
ydata.append([ 58.2822 , 0.4672 ])
ydata.append([ 64.8785 , 0.4693 ])
ydata.append([ 73.1584 , 0.4715 ])
ydata.append([ 83.8610 , 0.4738 ])
ydata.append([ 98.2316 , 0.4760 ])
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